

SkillSetu Sovereign National Platform

NATIONAL GOVERNANCE & AUDIT CONSOLE: ARCHITECTURE, ALGORITHMS, UX & BMC SPECIFICATION

Document Identifier:	SKILLSETU-DOC-ADMIN-2026-V3	Classification:	National Sovereign Audit Specification
Target Stakeholder:	Ministry of Education, AICTE, NITI Aayog, Regulators	Policy Alignment:	Digital India • DPDP Act 2023 • NEP 2020 • NCrF
Backend Engine:	Node.js 20 LTS + TypeScript + Express + Prisma	Audit AI Engine:	Google Gemini 2.5 Flash + Groq LPU Llama 3 70B

1. Executive Summary & National Governance Value Proposition

For government authorities, the Ministry of Education, AICTE, and national regulators, SkillSetu serves as **India's sovereign Digital Public Infrastructure (DPI) for human capital harmonization**. The Governance Console provides super-administrators with real-time macroeconomic telemetry on regional skill shortages, centralized registry management for universities and employers, automated fraud detection, and continuous audit verification of cryptographic TrustLedger blocks across the nation.

NATIONAL SECURITY & REGULATORY AUDIT ENCLAVE

Sovereign Audit & DPDP Act Compliance Enclave: The Governance Console enforces strict compliance with India's Digital Personal Data Protection (DPDP) Act 2023. Administrators have complete cryptographic visibility over data access audit trails, user consent timestamps, and affirmative action metrics, without compromising individual citizen privacy.

2. Technology Stack & Language Matrix for Governance Console

Layer / Tier	Technology / Framework	Technical Rationale & Architecture Role	Concrete Governance Benefit
Admin Terminal UI	React 18.3.1 + TypeScript + Tailwind CSS	High-security administrative console with role-restricted audit views and real-time telemetry tables.	Zero-compromise dashboard for national regulators with sub-second data filtration.
Role-Guard Security	React Router DOM 6.26 + ProtectedRoute	Strict cryptographic client-side and server-side guard enforcing allowedRoles: ['ADMIN'].	Absolute immunity against privilege escalation or unauthorized student/recruiter route access.
National Telemetry	Recharts 2.12.7 (SVG Compositor)	Country-wide skill deficit radars, placement volume curves, and institutional compliance matrices.	Actionable macro-economic charts for NITI Aayog policy formulation.
Server Architecture	Node.js 20 LTS + Express 4.19 (TypeScript)	Stateless REST APIs with hardened rate-limiting, CORS whitelisting, and strict request validation.	DDoS-resilient architecture capable of serving millions of concurrent national audit queries.
Database Layer	Prisma 5.17 ORM + SQLite / Turso	Centralized relational database maintaining the master registry of users, colleges, vacancies, and ledger hashes.	Complete referential integrity and auditability across millions of student records.
Predictive AI Engine	Google Gemini 2.5 Flash + Groq LPU Llama 3 70B	Dual-LLM forecasting pipeline evaluating nationwide industrial hiring trends against student enrollment.	Early warning alerts detecting emerging national talent deficits 1–3 years in advance.

3. Mathematical Algorithms & Regulatory Audit Engines

Administrative operations are governed by four advanced mathematical and macroeconomic intelligence algorithms:

Algorithm Name	Exact File Location	Mathematical / Technical Formulation	National Governance Advantage
1. National Skill-Shortage Forecasting Engine	services/aiService.ts (Lines 288–308)	Macroeconomic time-series regression cross-referencing national corporate job vacancy growth against university graduation cohorts to compute talent deficit percentage.	Empowers AICTE and NITI Aayog to allocate educational budgets and establish training centers ahead of critical industry shortages.

2. Merkle Root Cryptographic Chain Audit	controllers/studentFeatures.controller.ts & adminFeatures.controller.ts	Validates block continuity: checks $H_k \parallel \text{SHA-256}(H_{k-1} \parallel \text{Payload})$ across all historical attestation records.	Guarantees that no academic record, university grade, or corporate certificate has been tampered with or retroactively modified.
3. Cybercrime Job Fraud & Ghost Entity Scanner	services/aiService.ts (Lines 268–285)	Autonomous NLP risk scoring evaluating employer domain credentials, compensation-to-skill ratios, and illegal upfront fee demands.	Automatically quarantines suspicious recruiters, protecting vulnerable students from cyber financial fraud and fake employment rackets.
4. National Affirmative Action Telemetry	routes/admin.routes.ts & AdminDashboard.tsx	Statistical divergence index evaluating applicant selection rates across geographic tiers (Tier 1 vs Tier 2/3) and demographic categories.	Provides empirical proof that recruitment across the platform is merit-based and free from regional discrimination.

4. End-to-End Admin User Experience (UX) Blueprint

The administrative console provides complete oversight across institutional onboarding, opportunity audits, and system health:

Step	Interface / Route	Admin Action & Visual Feedback	Underlying System Telemetry
Step 1	Super-Admin Auth /login	Super-admin authenticates with multi-factor institutional credentials; system validates ADMIN role and routes to /admin.	Validates cryptographic session token and initializes audit logger.
Step 2	Governance Console /admin	Review nation-wide operational KPIs: Total Verified Students, Registered Universities, Active Corporate Opportunities, and Ledger Blocks.	High-performance aggregated count queries over central database tables.
Step 3	User & Entity Registry /admin/users	Search, filter, verify, or suspend student, recruiter, and academician accounts with instant role and verification status controls.	Executes paginated queries over users with role and institution filters.
Step 4	Opportunities Registry /admin/opportunities	Audit active corporate vacancies across India. Review fraud risk scores and de-list non-compliant or scam job postings with 1 click.	Updates opportunities.is_active status and dispatches recruiter notice.
Step 5	Application Audits /admin/applications	Monitor nation-wide placement flow, apprentice hiring conversion rates, and affirmative action benchmarks across all registered colleges.	Aggregated joins between applications, opportunities, and users.
Step 6	Skill Taxonomy Registry /admin/skills	Curate and standardize the national skill ontology aligned with NSQF Levels 4.5 through 7 and international classifications.	Manages master skills table and category mappings.
Step 7	Governance Dispatch /admin/notifications	Broadcast national policy notices, curriculum advisories, and system security announcements to all stakeholder dashboards.	Bulk insertion into notifications table for targeted user roles.

5. API Key Architecture, Endpoints & Security Encapsulation

API Key / Parameter	Declared Location	Server Ingestion File	Operational Usage & Security Guard
GEMINI_API_KEY	backend/.env	services/aiService.ts (Lines 6-35, 288-308)	Powers national skill-shortage early warning forecasting. Completely isolated on the backend server.
GROQ_API_KEY	backend/.env	services/aiService.ts (Lines 7, 42-69)	High-speed failover inference ensuring zero downtime for regulatory fraud detection and audit scans.
JWT_SECRET	backend/.env	middleware/auth.ts (Lines 8-18)	Enforces super-admin cryptographic isolation; rejects all requests lacking role === 'ADMIN'.

6. Societal Impact, National Policy & Demographic Dividend

SkillSetu serves as a national catalyst for India's 2047 economic vision (Viksit Bharat):

- **Harnessing India's Demographic Dividend:** With over 40 million students in higher education, SkillSetu optimizes workforce placement efficiency, transforming raw student potential into high-productivity GDP output.
- **Empirical Policy Data for NITI Aayog & AICTE:** Policy makers no longer rely on outdated sample surveys. The platform provides real-time, live macroeconomic data on talent surpluses and deficits across every state.
- **Digital Public Infrastructure (DPI) Leadership:** Built on the foundational philosophy of Aadhaar, UPI, and ONDC, SkillSetu establishes India as the global benchmark for sovereign credential attestation.
- **Protection Against Cyber Financial Crime:** Autonomous fraud detection stops fake recruitment agencies from extorting money from job-seeking youth.

7. Business Model Canvas (BMC) — National Governance & DPI Dimension

BMC Building Block	National Governance & DPI Strategic Implementation
Customer Segments	Ministry of Education, Ministry of Skill Development (MSDE), AICTE, UGC, State Higher Education Councils, NITI Aayog.
Value Propositions	Sovereign human capital intelligence, real-time national skill deficit telemetry, tamper-proof credential ledger, zero-bias placement oversight.
Channels	National Digital Public Infrastructure (DPI) portal, NIC Cloud (MeghRaj), state government API gateways, official AICTE dashboards.
Customer Relationships	Inter-ministerial governance boards, regulatory compliance audits, quarterly macroeconomic human capital reviews.
Revenue / Funding Streams	Digital India Public Grant Funding; Public-Private Partnership (PPP) Model; Enterprise Attestation Verification API Licensing: Commercial background verification firms pay per cryptographic ledger API query.
Key Resources	National competency ledger, sovereign skill ontology, predictive deficit models, inter-ministerial regulatory access controls.
Key Activities	Macroeconomic data aggregation, ledger cryptographic root verification, regulatory compliance monitoring, platform cybersecurity hardening.
Key Partnerships	Ministry of Education, AICTE, NITI Aayog, National Informatics Centre (NIC), NASSCOM, State Skill Development Missions.
Cost Structure	Government cloud infrastructure (MeghRaj / AWS GovCloud), national cybersecurity SOC monitoring, continuous algorithmic audit certification.